

General Guidelines for Presentation

General Guidelines for writing PPT and project Report: Both should have same content. Report being the detailed explanation of the PPT points that you explain orally while presenting.

- **PPT :**

- The content should be only the important points in the form of bullet list with meaningful short sentences and phrases. Running texts and big paragraphs are not accepted.
- It should have proper citations.
- Diagrams should be original and should be visible clearly.
- Use minimum font size of 17
- Put mathematical models and algorithms / methods explanations where ever necessary.
- Emphasize on your original thinking, original work and contribution and highlight them in the slide.

- **Project Report:**

- The PPT content and the project report content should match.
- All the points of PPT should be well explained and where needed they should be justified by using citations and /or by proper analysis and proofs.
- Emphasize on your original thinking, original work and contribution and highlight them in the appropriate places in the report.
- The diagrams / figures and tables should be numbered and tilted and should be refereed in the text with their numbers mentioning the purpose of the same. Also explain them in your own words w.r.t. the purpose, the contents, relation of various entities in them, etc.
- All references should be used as citations at the appropriate places.

The following chapter slides will give general information of the points to be considered and the sequence in which they need to be explained to give a proper flow to the report and to the presentation.

Nevertheless, you are free to modify, add or delete few points as per your problem needs.



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DataSentinel – Data Quality Intelligence System

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Project Guide
Ms. Priyanka Karhade

Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Sustainable Development Goals (SDG) mapped

- ▶ SDG 9: Industry, Innovation and Infrastructure
 - ▶ Mapping: Builds resilient infrastructure by democratizing data intelligence through a scalable, automated AI pipeline.
- ▶ SDG 4: Quality Education
 - ▶ Mapping: Serves as an interactive learning platform for students and analysts to understand data quality without coding prerequisites.



Introduction

- **Real-time Observation:** Modern datasets are growing exponentially large and complex.
- Current Issue: Applying machine learning or anomaly detection without proper exploration causes unreliable insights and biased modeling.
- **Motivation:** Existing systems rely heavily on manual, unscalable, and generic cleaning methods.
- **Objectives:** Design a context-aware data preprocessing framework.
 - Develop a query-driven data interaction layer.
 - Implement an explainable anomaly detection module.
- Original Contribution: Democratizing data intelligence via an integrated, no-code, multi-algorithm platform with quantitative quality scoring and natural language interfaces.

Literature Survey of the existing system

Sr. No .	Title	Author (s)	Year	Outcomes	Methodology	Result
1.	[1]Are Large Language Models Ready for Multi-Turn Tabular Data Analysis?	J. Li, N. Huo, et al.	2025	CoTA benchmark and Adaptive Conversation Reflection (ACR).	Uses a multi-agent sandbox to evaluate LLMs on 1013 conversational data tasks.	Result: ACR improved Claude-3.5-Sonnet by 35.14%. Limitation: Focuses on evaluation metrics, not a deployable no-code cleaning tool.
2.	[2]TiInsight: A SQL-based Automated EDA System through LLMs	Jun-Peng Zhu, et al.	2026	TiInsight, a cross-domain Exploratory Data Analysis system.	Uses LLMs for Hierarchical Data Context generation, Text-to-SQL, and visualization.	Result: 86.3% Text-to-SQL accuracy on Spider dataset. Limitation: Strictly SQL-driven; lacks ML models for intelligent anomaly detection.

Literature Survey of the existing system

Sr. No .	Title	Author (s)	Year	Outcomes	Methodology	Result
1.	[3]A ML Framework for Automated Data Cleaning & Anomaly Detection	Jitendra Agrawal, et al.	2025	Automated pipeline for data cleaning and error imputation.	Integrates Isolation Forests (anomalies), PCA (dimensionality), and Random Forests (imputation).	Result: Achieved 0.88 precision and 0.015 Mean Absolute Error. Limitation: Lacks a Natural Language Interface (NLP) and explainability for non-technical users.

Limitations of existing systems

- **Fragmented Solutions:** Current systems either focus solely on NLP querying (lacking ML cleaning) or strictly on ML processing (lacking user-friendly interaction).
- **High Technical Barrier:** Robust data cleaning frameworks still require programming expertise and manual intervention.
- **Lack of Explainability (XAI):** Existing ML models flag anomalies but fail to explain why a data point is anomalous in a transparent format.
- **Justification for DataSentinel:** We are building a unified, no-code platform that integrates statistical ML analysis with LLMs, solving both the technical barrier and the explainability gap.

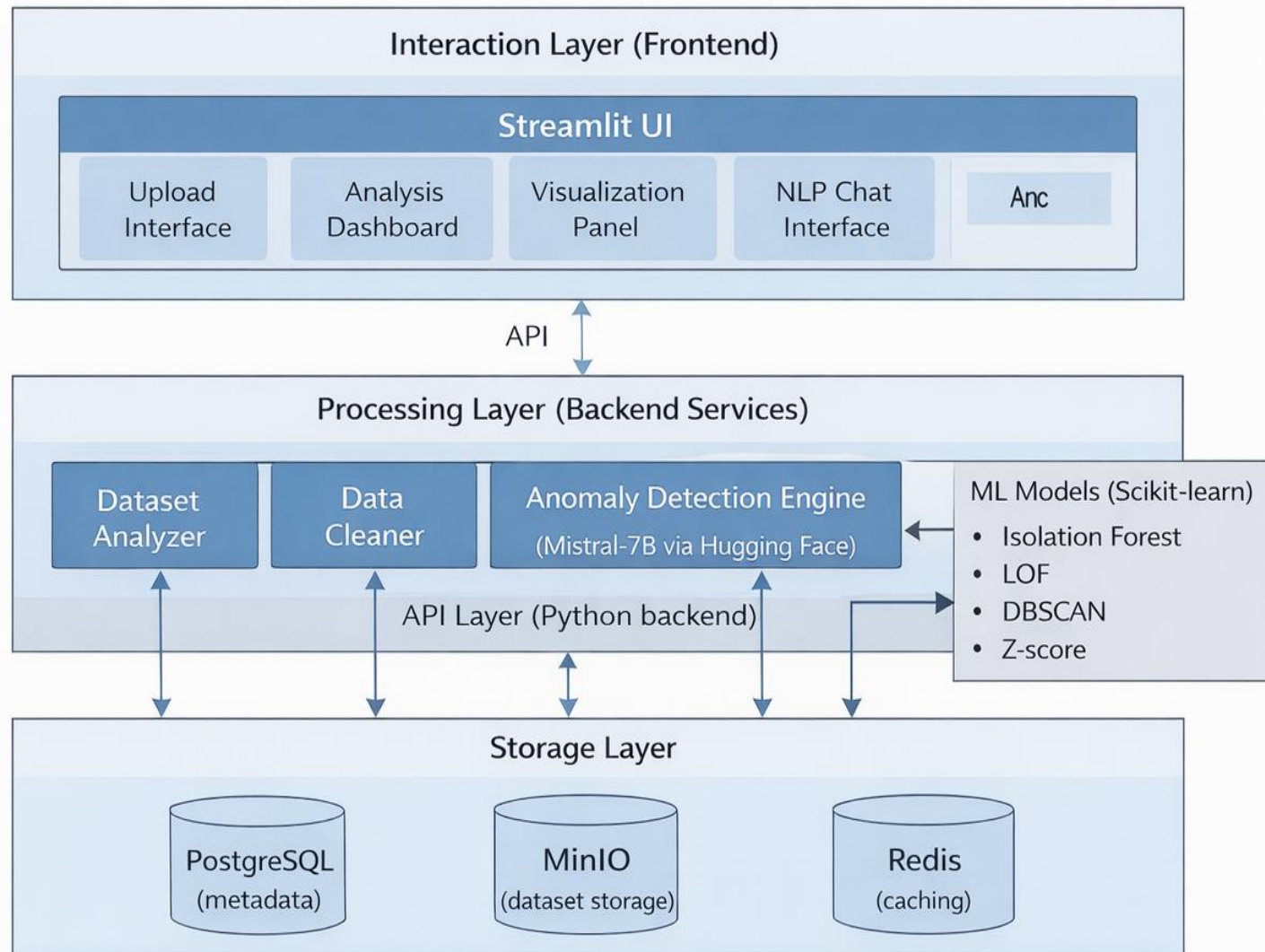
Problem statement

- **Core Problem:** Modern datasets are highly complex. Applying advanced modeling without proper data exploration leads to biased outcomes and unreliable insights.
- **The Gap:** Users lack a unified, scalable system that seamlessly bridges the gap between raw data upload and intelligent, context-aware analysis.
- **Proposed Architecture:** DataSentinel, an AI-powered platform combining statistical profiling and Natural Language Processing.
- **Key Interventions (Our Contribution):**
 - Implements domain-informed cleaning strategies using Pandas and Scikit-learn.
 - Provides a query-driven interaction layer using natural language and SQL.
 - Uses a multi-algorithm ensemble (Isolation Forest, LOF) to generate explainable anomaly confidence scores.

System Design - Architecture Overview

- **Modular Monolith Architecture:** System is divided into independent, scalable layers (Frontend, Backend Services, and Storage).
- **Frontend Layer (Interaction):** Built for no-code interaction, featuring upload, analysis, and visualization modules.
- **Backend Services (Processing):** Houses the core engines: DatasetAnalyzer, DataCleaner, AnomalyDetector, and NLPChatbot.
- **Storage Layer (Persistence):** Utilizes MinIO for dataset storage, Redis for caching, and PostgreSQL for metadata.
- **Original Contribution:** A seamless pipeline where natural language queries directly trigger complex ML backend services without user coding.

System Design - Architecture Overview



System Design - Operational Workflow

- Data Ingestion & Validation: Automated format detection (CSV, Excel, Parquet) and loading into memory-efficient DataFrames.
- Intelligent Profiling: The DatasetAnalyzer executes statistical profiling, missing value analysis, and regex-based pattern recognition.
- Intelligence & Recommendations: Generates AI-powered summaries and calculates data quality scores with severity rankings.
- Interactive Processing: Users execute domain-informed cleaning transformations with real-time before/after validation.
- Explainable Anomaly Detection: Applies an ensemble of algorithms, generating visual score distributions and human-readable explanations.

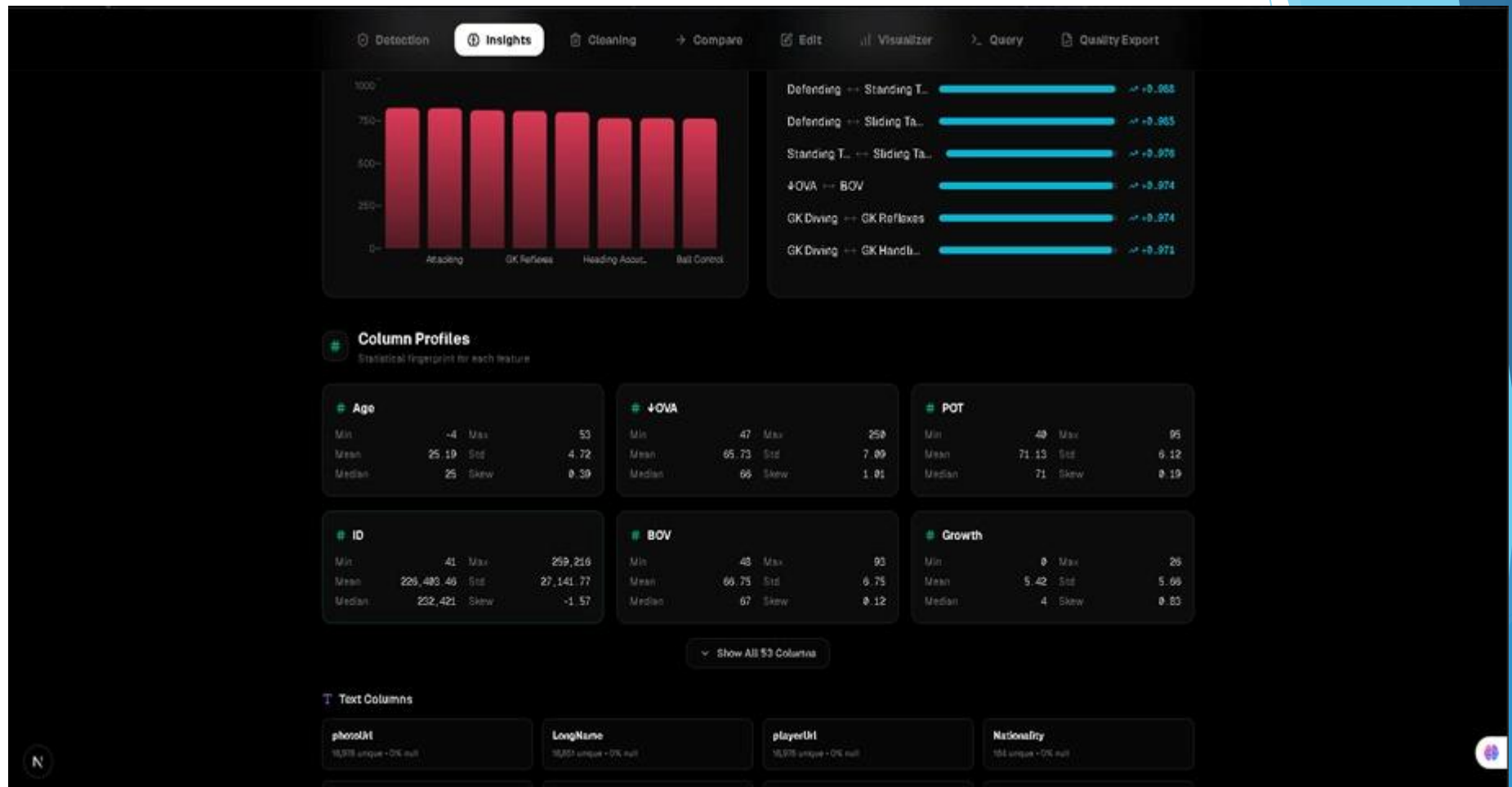
Technologies

- Frontend & Visualization:
 - Streamlit: Chosen for rapid, data-centric web prototyping.
 - Plotly & Seaborn: For interactive, scalable statistical visualizations.
- Data Processing & Backend:
 - Pandas & NumPy: Core engines for numerical computing and data manipulation.
 - SciPy: Utilized for statistical tests and distributions.
 - AI & Machine Learning Infrastructure:
 - Scikit-learn: Foundation for anomaly detection algorithms.
 - Hugging Face API & Mistral-7B: Powers the natural language conversational interface and automated code generation.
- DevOps & Storage: Docker (Containerization), PostgreSQL, and Redis.

Methodologies & Algorithms

- Ensemble Anomaly Detection (Original Approach):
 - Isolation Forest: Tree-based ensemble method optimized for high-dimensional dataset outliers.
 - Local Outlier Factor (LOF): Density-based clustering to detect local anomalies in varying density regions.
 - DBSCAN: Combined for spatial clustering and classical statistical flagging.
- Explainable AI (XAI) Methodology: Translates mathematical anomaly scores into human-readable text explanations for end-users.
- Incremental Processing Methodology: Ensures memory-efficient handling of large datasets by chunking processing tasks.
- Natural Language Processing: Uses LLMs for context-aware Q&A and regex for strict pattern recognition (emails, URLs).

Implementation



Conclusion

- Democratizing Data Intelligence: DataSentinel successfully bridges the gap between raw, complex datasets and ML-ready data without requiring programming expertise.
- Enhanced Trust via XAI: By providing explainable outputs for anomaly detection, the system builds user trust in automated data cleaning.
- Scalable and Robust: The integration of statistical profiling, multi-algorithm ML, and LLM-driven interaction creates a comprehensive, future-proof EDA platform.
- Goal Achieved: Successfully designed a context-aware framework that significantly reduces the time and error rate associated with manual data preparation.

References

- [1]Jun-Peng Zhu, et al., "TiInsight: A SQL-based Automated Exploratory Data Analysis System through Large Language Models", arXiv preprint arXiv:2601.09404, 2026.
[Link: https://arxiv.org/abs/2601.09404](https://arxiv.org/abs/2601.09404)
- [2]J. Li, N. Huo, et al., "Are Large Language Models Ready for Multi-Turn Tabular Data Analysis?", arXiv preprint arXiv:2502.13066, 2025.
[Link: https://arxiv.org/abs/2502.13066](https://arxiv.org/abs/2502.13066)
- [3]Jitendra Agrawal, et al., "A Machine Learning Framework for Automated Data Cleaning and Anomaly Detection in Large Datasets", Journal of Data Science and Analytics, vol. 12, pp. 45-62, 2025.
[Link: https://doi.org/10.26438/ijsrcse.v13i3.679](https://doi.org/10.26438/ijsrcse.v13i3.679)

Thank You...!!